SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA0510** **Slot A**

COURSE OUTCOME COVERED
CO3: ***Analyze and extract image features using detection and matching techniques for effective image representation and comparison.*** **(BTL 4)**
Assessment Tool Weightage: 10%

QUESTIONS:50 **Total Marks 50** **Duration :60 Minutes**
Annexure A MCQ

Code of Conduct:

I M.Manoj Kumar(1924225114)____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **M,Manoj Kumar**

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	*Marks(100)*
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Preprocessing – Normalization & Scaling

A 12-bit image has pixel value r = 3000. Normalized to [0,1], scaled to 8-bit [0,255]. Final pixel value?

A) 150

B) 187

C) 200

D) 225

Answer: B) 187

2. Image Storage Calculation

RGB image 2048×1024, 8 bits/channel. Storage in MB?

A) 6 MB

B) 8 MB

C) 12 MB

D) 24 MB

Answer: A) 6 MB

3. Sobel Edge Magnitude

Gx=10, Gy=24. Gradient magnitude (approx)?

A) 20

B) 25

C) 26

D) 30

Answer: C) 26

4. Sobel Edge Direction

Gx=10, Gy=24. Edge direction in degrees?

A) 22°

B) 67°

C) 78°

D) 90°

Answer: B) 67°

5. Harris Corner Response

det(M)=500, trace(M)=40, k=0.04. Compute R.

A) 440

B) 460

C) 500

D) 520

Answer: A) 440

6. Harris Corner vs Edge Classification

Eigenvalues $\lambda 1=100$, $\lambda 2=5$. Pixel is:

A) Flat region

B) Edge

C) Corner

D) Noise

Answer: B) Edge

7. Hough Transform – Line Parameter

$\rho=50$, $\theta=90°$. Which line is represented?

A) Horizontal line y=50

B) Vertical line x=50

C) Diagonal

D) Random line

Answer: A) Horizontal line y=50

8. SIFT Scale Adjustment

$\sigma=1.2$, image scaled by factor 2. New keypoint scale?

A) 1.2

B) 2.4

C) 0.6

D) 3.2

Answer: B) 2.4

9. PCA Variance Calculation

Eigenvalues: 50,25,15,10. Components for ≥90% variance?

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

10. PCA Dimensionality Reduction Impact

200 dims reduced to 40. Percentage reduction?

A) 70%

B) 75%

C) 80%

Answer: C) 80%

11. Preprocessing – Normalization Scenario

12-bit image r=2500. Normalized value?

A) 0.55

B) 0.61

C) 0.65

D) 0.72

Answer: B) 0.61

12. Image Storage Scenario

2048×1024 grayscale 16-bit → 8-bit. % reduction in memory?

A) 25%

B) 50%

C) 60%

D) 75%

Answer: B) 50%

13. Edge Detection Scenario

Gx=12, Gy=16. Gradient magnitude?

A) 18

B) 20

C) 24

D) 28

Answer: B) 20

14. Edge Direction Scenario

Gx=12, Gy=16. Gradient direction θ?

A) 36.9°

B) 45°

C) 53.1°

D) 60°

Answer: A) 36.9°

15. Harris Corner Detection Scenario

det(M)=600, trace(M)=30, k=0.05. Compute R.

A) 540

B) 555

C) 570

D) 600

Answer: B) 555

16. Harris Corner Classification Scenario

$\lambda 1=100$, $\lambda 2=90$. Pixel is most likely:

A) Flat region

B) Edge

C) Corner

D) Noise

Answer: C) Corner

17. Hough Transform Scenario

Curves intersect at $\rho=80$, $\theta=60°$. Indicates?

A) Detected corner

B) Line in image space

C) Circular feature

D) Noise

Answer: B) Line in image space

18. SIFT Scale Scenario

$\sigma=2.0$, image downscaled by 0.5. New keypoint scale?

A) 0.5

B) 1.0

C) 2.0

D) 4.0

Answer: B) 1.0

19. PCA Variance Scenario

Eigenvalues: 70,15,10,5. Components for ≥90% variance?

A) 2

B) 3

C) 4

Answer: B) 3

20. PCA Dimensionality Reduction Scenario

150 dims reduced to 25. % reduction?

A) 70%

B) 75%

C) 83%

D) 85%

Answer: C) 83%

21. Preprocessing – Normalization & Scaling

12-bit satellite image r=3500. Final 8-bit pixel value?

A) 190

B) 210

C) 218

D) 225

Answer: C) 218

22. Image Storage Analysis

1024×1024 RGB 8-bit → grayscale 8-bit. % reduction in storage?

A) 33%

B) 50%

C) 66%

D) 75%

Answer: C) 66%

23. Edge Detection – Gradient Magnitude

Gx=14, Gy=48. Gradient magnitude?

A) 46

B) 50

C) 50.5

D) 50.99

Answer: B) 50

24. Edge Direction

Gx=14, Gy=48. Edge orientation in degrees?

A) 70°

B) 73°

C) 75°

D) 77°

Answer: B) 73°

25. Harris Corner Response

det(M)=400, trace(M)=35, k=0.05. Compute R.

A) 320

B) 350

C) 378

D) 400

Answer: B) 350

26. Harris Corner Classification

$\lambda 1=130$, $\lambda 2=120$. Pixel type?

A) Flat

B) Edge

C) Corner

D) Noise

Answer: C) Corner

27. Hough Transform – Line Detection

Curves intersect at $\rho=100$, $\theta=45°$. Indicates?

A) A line passes through both points

B) A corner formed by lines

C) Circular feature

D) Random noise

Answer: A) A line passes through both points

28. SIFT Scale Scenario

σ=1.5, image scaled by 0.25×. New keypoint scale?

A) 0.375

B) 0.75

C) 1.5

D) 6

Answer: A) 0.375

29. PCA Variance Retention

Eigenvalues: 60,20,10,5,5. Min components for ≥90% variance?

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

30. PCA Dimensionality Reduction

180 dims → 30 components. % reduction?

A) 70%

B) 75%

C) 83%

D) 85%

Answer: C) 83%

31. Preprocessing Method Selection

High sensor noise, uneven illumination; enhance edges. Best combination?

A) Histogram equalization + median filtering

B) Gaussian smoothing + normalization

C) Median filtering + adaptive histogram equalization

D) Low-pass filtering + quantization

Answer: C) Median filtering + adaptive histogram equalization

32. Image Representation Trade-Off

1024×1024 medical scan: 16-bit grayscale vs 8-bit with HE. Best for detail & storage?

A) 16-bit grayscale

B) 8-bit grayscale with HE

C) 8-bit grayscale without HE

D) RGB 24-bit

Answer: B) 8-bit grayscale with HE

33. Edge Detection Algorithm Choice

Satellite image, low contrast edges, some noise. Most suitable detector?

A) Sobel

B) Prewitt

C) Canny

D) Roberts

Answer: C) Canny

34. Harris vs FAST Corner Selection

Real-time object tracking. Preferred for speed without losing accuracy?

A) Harris

B) FAST

C) SIFT

D) Hough Transform

Answer: B) FAST

35. Hough Transform Line Detection Evaluation

Method 1 (1px,1°) vs Method 2 (5px,5°). Best balance of accuracy/cost?

A) Method 1

B) Method 2

C) Both equal

D) Use Canny edges only

Answer: A) Method 1

36. SIFT vs SURF Evaluation

Matching keypoints with scale & rotation differences; robust + lower computation?

A) SIFT

B) SURF

C) Harris + SSD

D) Hough Transform

Answer: B) SURF

37. PCA Component Selection

Eigenvalues: 70,20,10,5,5. Components to retain ≥95% variance?

A) 2 components

B) 3 components

C) 4 components

D) 5 components

Answer: C) 4 components

38. Feature Extraction Evaluation

Noisy industrial image; detect straight edges. Optimal combination?

A) Sobel + Hough Transform

B) Canny + Harris Corner

C) Median Filter + SIFT

D) Gaussian Blur + PCA

Answer: A) Sobel + Hough Transform

39. Preprocessing for SIFT Matching

Aerial images with different brightness/contrast. Best preprocessing for scale-invariant matching?

A) Histogram equalization + normalization

B) Median filtering + low-pass filter

C) Gaussian smoothing only

D) No preprocessing

Answer: A) Histogram equalization + normalization

40. PCA Dimensionality Evaluation

200 dims → 30 components (~90% variance); need fast classification. Correct evaluation?

A) Reduction sufficient for speed and accuracy

B) Reduce to 10 components for maximum speed

C) Keep 200 components for highest accuracy

D) Use SIFT only

Answer: A) Reduction sufficient for speed and accuracy

41. Preprocessing – Normalization & Decision

12-bit satellite image r=3200. Final pixel value?

A) 198

B) 199

C) 200

Answer: B) 199

42. Image Storage Trade-Off

1024×1024 RGB 8-bit → grayscale 8-bit. % reduction?

A) 33%

B) 50%

C) 66.7%

D) 75%

Answer: C) 66.7%

43. Sobel Edge Magnitude Evaluation

Gx=12,Gy=16 vs Gx=10,Gy=18. Magnitude & comparison?

A) 20; weaker

B) 20; stronger

C) 21; weaker

D) 21; stronger

Answer: A) 20; weaker

44. Harris Corner Response Analysis

det(M)=400,trace=30,k=0.04 vs det=350,trace=25,k=0.04. R & likelihood?

A) R=364; less likely

B) R=364; more likely

C) R=362; less likely

D) R=362; more likely

Answer: B) R=364; more likely

45. Harris Eigenvalue Evaluation

λ1=120, λ2=80. Classify pixel numerically.

A) Flat

B) Edge

C) Corner

D) Noise

Answer: C) Corner

46. Hough Transform Voting Numerical

Points (100,50) & (200,100) intersect at $\rho=70,\theta=45°$. Indicates?

A) Line passes through both points

B) Corner

C) Noise

D) Circle

Answer: A) Line passes through both points

47. SIFT Keypoint Scale Numerical Evaluation

$\sigma=1.6$, image scaled 0.5×. New σ?

A) 0.5

B) 0.8

C) 1.6

D) 3.2

Answer: B) 0.8

48. PCA Variance Contribution Evaluation

Eigenvalues: 60,20,10,5,5. Min components for ≥90% variance?

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

49. PCA Dimensionality Reduction Numerical Evaluation

200 dims → 30 components. % reduction?

A) 70%

B) 75%

C) 80%

D) 85%

Answer: C) 80%

50. Preprocessing Impact Evaluation

Image A (uniform illumination) vs Image B (low contrast/brightness diff), both get HE. Which benefits more?

A) Image A

B) Image B

C) Both equal

D) Neither

Answer: B) Image B